

Project Progress Report: Cross-Domain Generalization with MNIST and Fashion-MNIST

Team Members: Ziyue Zhao, Barbara Kotlan, Zoha Ahmed, Xiangnan Zhang

1. Overview of Project Goals

This project investigates the problem of domain shift in deep learning by evaluating how models trained on one dataset (MNIST) generalize to a different but structurally similar dataset (Fashion-MNIST). We aim to quantify performance degradation and assess whether domain adaptation techniques—CORAL, DANN, Bayesian Neural Networks (MC Dropout), and IRM—can mitigate this gap.

2. Progress to Date (Up to Week 6)

2.1 Data Preparation and Baseline Model

We have successfully completed the initial project setup, including dataset acquisition and preprocessing for both MNIST and Fashion-MNIST. A baseline Convolutional Neural Network (CNN) architecture has been implemented and trained on the MNIST dataset.

Preliminary evaluation shows:

- High accuracy on MNIST (source domain), indicating correct model implementation.
- Significant performance drop on Fashion-MNIST (target domain), confirming the presence of domain shift.

These results establish a strong baseline for evaluating domain adaptation methods.

2.2 Implementation of Domain Adaptation Methods

Progress on the four main approaches is as follows:

- CORAL (Correlation Alignment)
The CORAL loss has been integrated into the CNN training pipeline. Initial experiments show partial alignment of feature distributions, with modest improvements in cross-domain accuracy compared to the baseline.
- DANN (Domain-Adversarial Neural Network)
The adversarial framework, including gradient reversal and domain classifier, has been implemented. Preliminary runs indicate that the model is beginning to learn domain-invariant features, although training stability and hyperparameter tuning are ongoing challenges.
- Bayesian Neural Network (MC Dropout)
Monte Carlo Dropout has been incorporated into the CNN architecture. Initial

experiments successfully produce uncertainty estimates, allowing us to distinguish between confident and uncertain predictions on target-domain data.

- IRM (Invariant Risk Minimization)

The IRM penalty has been implemented and integrated into training. Early results suggest that enforcing invariance constraints is computationally more demanding, and tuning the penalty weight is critical for achieving meaningful improvements.

2.3 Preliminary Results and Observations

From initial experiments:

- All methods show some improvement over the baseline, but to varying degrees.
- DANN and CORAL appear most promising in improving cross-domain accuracy so far.
- Bayesian approaches provide valuable uncertainty estimates but do not directly optimize accuracy.
- IRM shows potential but requires careful tuning and more stable training.

Additionally, we have begun generating:

- Confusion matrices to analyze class-wise performance
- t-SNE visualizations to inspect feature alignment across domains
- Uncertainty plots for Bayesian model outputs

3. Challenges Encountered

- Training instability in DANN due to adversarial optimization
- Hyperparameter sensitivity across all methods (e.g., CORAL loss weight, IRM penalty)
- Computational overhead, especially for IRM and MC Dropout (multiple forward passes)
- Difficulty in fair comparison across methods without consistent tuning

4. Next Steps (Weeks 6–9)

To complete the project, we plan to focus on the following:

4.1 Experimentation and Evaluation

- Conduct systematic hyperparameter tuning for all methods
- Run controlled experiments to ensure fair comparison
- Perform ablation studies to understand the contribution of each component

4.2 Analysis and Visualization

- Finalize t-SNE plots for feature representation comparison
- Generate comprehensive confusion matrices

- Analyze uncertainty calibration for Bayesian models

4.3 Comparative Study

- Compare all methods using:
 - Cross-domain accuracy
 - Robustness to domain shift
 - Computational efficiency
- Identify trade-offs between performance and complexity

4.4 Final Deliverables

- Compile results into structured tables and figures
- Write the final report, including methodology, results, and discussion
- Prepare presentation slides summarizing key findings

5. Current Status Summary

At this stage, the project is on track with the proposed timeline. The baseline and all four domain adaptation methods have been implemented, and preliminary experiments have been conducted. The focus now shifts toward rigorous evaluation, deeper analysis, and clear presentation of results.

This progress establishes a solid foundation for completing the project and achieving a comprehensive comparison of domain adaptation techniques for cross-domain generalization.